

A vortex lattice method with ventilation: methods, usage and validation

Repository `Fluid-Dynamics-Laboratory/vlm-ventilation`, branch `vent/integration`. September 2026.

This document describes how the code works, how to run it, which measurements validate it and how well it does against them. It is written for an engineering undergraduate who has followed a first course in aerodynamics and knows what a vortex, a boundary layer and a lift coefficient are. Section 1 explains the problem and the plan of the code. Sections 2 to 8 are the methods, in the order in which the code executes them. Section 9 shows one complete run. Section 10 is the validation: which measured curves test the code, how the code compares with the ones digitised so far, and what remains to be digitised. Section 11 lists what the code does not do.

Every quantity is nondimensional unless a unit is stated. Symbols with a hat, such as $\hat{c}$, are dimensional; the code itself works in metres, seconds and radians.

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1 The problem and the plan of the code

A hydrofoil that pierces the free surface, or runs close beneath it, may *ventilate*: air from the atmosphere is drawn into the low-pressure separated flow on its suction side and forms a cavity. When the cavity covers the suction side the lift falls by 50 to 70 % (Harwood, Young and Ceccio, 2016). For a foiling boat this is a sudden loss of the force that holds the hull out of the water. The code predicts, for a given foil, speed and immersion, (i) the loads when the flow is fully wetted, (ii) whether ventilation incepts and by which route, (iii) the loads once a cavity is present, and (iv) whether the cavity persists or washes out when the conditions change, which gives the hysteresis observed in experiments.

The code is a **vortex lattice method** (VLM), a potential-flow method in which the lifting surface is replaced by a lattice of vortex rings, with a free wake that is shed and convected in time. It was developed during an internship

(Orain, 2026) and revised in this work. Three layers sit on top of it (Fig. 1.1).

- **Section data** (Sect. 5): two-dimensional viscous properties of the foil section, from a surrogate of XFOIL, read as a table. The lattice has no boundary layer, so separation and the true suction peak come from here.
- **Inception model** (Sect. 6): reads a converged wetted solution and decides, section by section, whether ventilation can start and by which route.
- **Cavity model and regime machine** (Sects. 7 and 8): once ventilation has started, a cavity at atmospheric pressure is added to the lattice, its effect on the loads is solved, and a state machine decides whether the cavity persists or washes out.

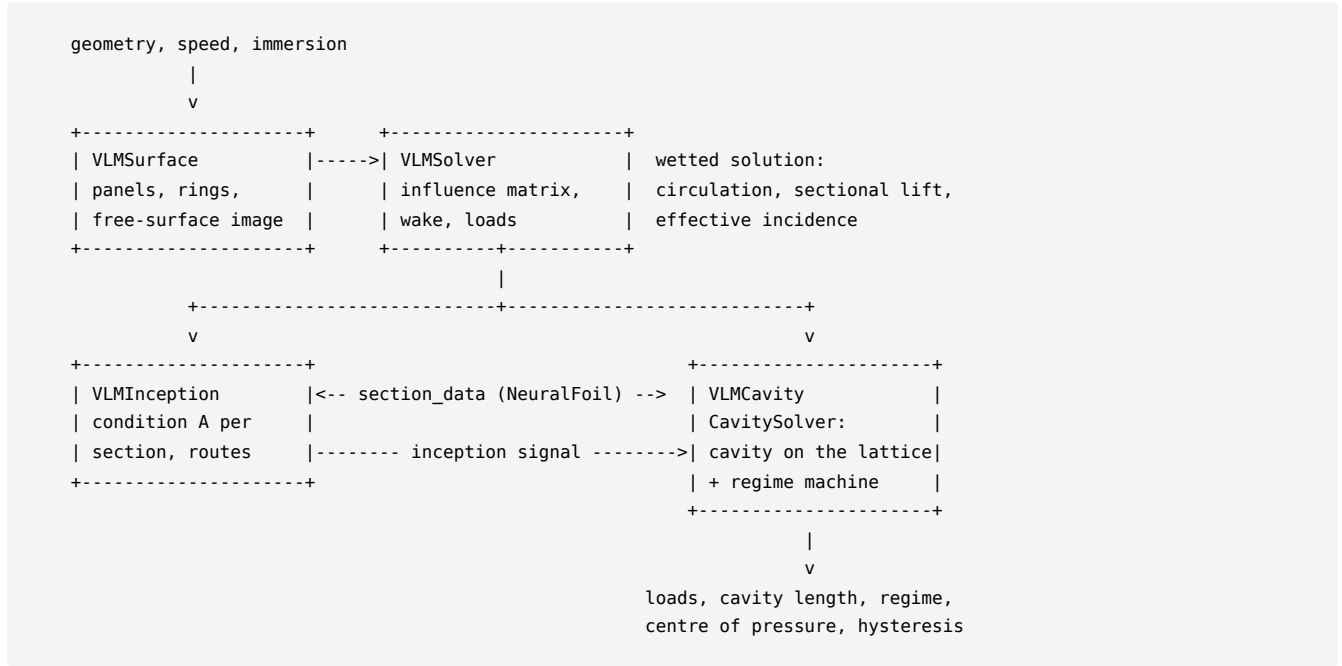


Figure 1.1. Data flow of the code. Each box is one Python module in `src/`.

The layers are independent. The inception model reads a solution and writes a dictionary; the cavity model accepts any function that produces such a dictionary. This is why the two were developed on separate branches (`vent/inception`, `vent/cavity-regime`) and are merged here.

2 Nomenclature

Symbol	Meaning
α	geometric incidence of the surface; for a vertical strut it is the drift angle
α_{eff}	effective incidence of a section, $C_l/(2\pi \cos \lambda)$
α_{sep}	incidence at which the section's suction side separates (stall)
a_0	sectional lift slope; 2π for a thin wetted section
AR, AR_h	aspect ratio; immersed aspect ratio $h/\bar{c}$ of a surface-piercing strut
b, b_{min}	span; smallest panel dimension of the lattice
$c, \bar{c}$	local chord; mean chord
C_L, C_l	lift coefficient of the surface, on $\frac{1}{2}\rho u_\infty^2 S$; sectional lift coefficient

Symbol	Meaning
$C_p, C_{p,\min}$	pressure coefficient; its minimum on the suction side
d	cutoff distance of the Biot–Savart law
e	centre of pressure forward of mid-chord, in chords
Fn_h, Fn_c	depth Froude number $u_\infty/\sqrt{gh}$; chord Froude number $u_\infty/\sqrt{gc}$
g	gravitational acceleration
h	immersion of a strut (depth of its tip)
L	cavity length in chords
M, N	number of chordwise and spanwise panels
q	source strength of a cavity panel (transpiration velocity)
r_c	vortex core radius
Re_c	chord Reynolds number $u_\infty c/\nu$
u_∞	free-stream speed
Γ, γ	circulation of a ring; bound vorticity per unit chord
λ	mid-chord sweep angle
σ_c	cavitation number of an atmospheric cavity at depth z , $2g z /u_\infty^2$
ϕ	angle of the cavity closure line from the flow direction
Ψ	cavity parameter $\sigma_c/(2\alpha_{\text{eff}})$
FW, PV, FV	fully wetted, partially ventilated, fully ventilated regimes

Coordinates: x streamwise, z vertical with the undisturbed free surface at $z = 0$. A horizontal wing lies in the x – y plane (`plan=1`); a vertical strut lies in the x – z plane (`plan=0`) and occupies $0 \leq z \leq h$ below the surface.

3 Running the code

3.1 Installation

Python 3.10 or later. From the repository root:

```
pip install -r requirements.txt
```

This installs NumPy, Numba (which compiles the Biot–Savart kernel), Matplotlib, Jupyter and NeuralFoil. Nothing needs a compiler.

3.2 First run

The notebook `vlm_unsteady_wake.ipynb` runs the wetted solver on a rectangular wing of aspect ratio one at 10° and plots the wake and the sectional lift. From the repository root:

```
jupyter lab vlm_unsteady_wake.ipynb
```

The lift coefficient printed at the end should be 0.386. The same case as a script, in about five seconds:

```

import sys; sys.path.append("src")
import numpy as np
from VLMSurface import VLMSurface
from VLMSolver import VLMSolver

N, M = 15, 5 # spanwise panels per half-span, chordwise panels
wing = VLMSurface(origin=np.zeros(3), plan=1, boundary=False, shedding="both", b=1.0,
                  c=np.ones(N + 1), alpha=np.deg2rad(10), beta=0, lamb=0, delta=0, phi=0,
                  sym=True, space=True, n=N, m=M)

wing._build_wing()
vlm = VLMSolver([wing], u_inf=np.array([1.0, 0, 0]), boundary=False, ratio=0.05, a_ratio=0.4)
vlm._time_sim(t=3.0, dt=0.1, distribution="classic") # three chords of travel
vlm._kuttas_loads()
print("CL =", -2*wing.loads[2]) # 0.386

```

The arguments of `VLMSurface` are, in order: origin of the surface; plane (0 vertical, 1 horizontal); whether a free-surface image is built; which tips shed vorticity ("both" , "left" , "right" , "none"); span; chord at each of the $N + 1$ spanwise stations; incidence, drift, sweep, dihedral and twist in radians; whether the wing is mirrored about its root; cosine spacing towards the tips; and the panel counts. The two numbers passed to `VLMSolver` are the vortex core radius in chords and the cutoff of the boundary condition in units of the smallest panel dimension (Sect. 4.6).

3.3 Tests, verification and validation scripts

Command	What it does	Time
<code>python tests/test_inception.py</code>	unit checks of the section data and the inception routes	1 min
<code>python tests/test_cavity.py</code>	unit checks of the source kernel, the cavity and the regime machine	3 min
<code>python docs/tip_loading/study.py</code>	mesh and time-step convergence of the wetted solver, original against revised	3 min
<code>python docs/cavity/verify_acosta.py</code>	lattice cavity against Acosta's linear theory	5 min
<code>python docs/inception/validate.py</code>	inception against published values	25 min
<code>python docs/cavity/validate.py</code>	ventilated loads, closure angle and hysteresis on the strut of Harwood et al.	20 min
<code>python docs/report/example_end_to_end.py</code>	the complete run of Sect. 9	5 min
<code>python docs/report/build_html.py --pdf</code>	rebuilds this document as HTML and PDF from the Markdown source	1 min
<code>python docs/validation/import_digitised.py</code>	converts the raw digitised curves of <code>docs/validation/data/raw/</code> into the template format	1 s
<code>python docs/validation/validate.py</code>	compares the code with every digitised curve in <code>docs/validation/data/</code> (Sects. 10.2 to 10.5)	25 min

Each script prints its results and writes a JSON file next to itself. The first run of any script takes an extra minute while Numba compiles the kernel.

3.4 Repository layout

src/VLMPanel.py, VLMSurface.py, VLMSolver.py	the vortex lattice method (Sect. 4)
src/section_data.py, section_data/*.json	section data (Sect. 5)
src/VLMInception.py	inception model (Sect. 6)
src/VLMCavity.py, vent_section.py	cavity model and regime machine (Sects. 7, 8)
tests/	unit checks
docs/tip_loading/	convergence study and its report (Sect. 4.7)
docs/inception/, docs/cavity/	per-model documentation and validation scripts
docs/report/	this document (Markdown, HTML and PDF), its figures, the complete run
docs/validation/	digitised data, the importer and the comparison script (Sect. 10)
vlm_unsteady_wake.ipynb	the working notebook

4 The vortex lattice method

4.1 Idea

In potential flow a thin lifting surface can be replaced by a sheet of vorticity. The vortex lattice method discretises the sheet into $N \times M$ quadrilateral panels, each carrying a closed vortex ring of unknown circulation Γ_k (Katz and Plotkin, 2001, Sect. 12.3). Every ring induces a velocity everywhere in the fluid by the Biot–Savart law. The rings' circulations are found from one condition per panel: the flow must not cross the surface at the panel's control point. Once the circulations are known, the loads follow from the Kutta–Joukowski theorem.

4.2 Geometry and panels (VLMSurface , VLMPanel)

The planform is built from the chord at each spanwise station, the sweep, dihedral and twist. Spanwise stations are spaced uniformly or clustered towards the tips with a cosine law, which resolves the tip loading. Each panel has four corners; its vortex ring is the panel shifted downstream by a quarter of the panel chord, so that the ring's leading segment, the *bound vortex*, lies at the panel's quarter chord and the control point at its three-quarter chord. This placement makes a single panel satisfy the two-dimensional Kutta condition exactly (Pistoletti's theorem).

A symmetric wing is built from one half and mirrored. A surface-piercing strut is built with `plan=0`, its span along z , and `sym=False`, so that it occupies $0 \leq z \leq h$ with the free surface at $z = 0$.

4.3 Influence coefficients and the boundary condition (VLMSolver._build_A , _build_b)

The velocity induced at a point by a straight vortex segment of circulation Γ from $\mathbf{r}_1$ to $\mathbf{r}_2$ is

$$\mathbf{v} = \frac{\Gamma}{4\pi} \frac{\mathbf{r}_1 \times \mathbf{r}_2}{|\mathbf{r}_1 \times \mathbf{r}_2|^2} \mathbf{r}_0 \cdot \left(\frac{\mathbf{r}_1}{|\mathbf{r}_1|} - \frac{\mathbf{r}_2}{|\mathbf{r}_2|} \right), \quad \mathbf{r}_0 = \mathbf{r}_2 - \mathbf{r}_1. \quad (4.1)$$

The kernel `_induced_velocity` evaluates (4.1) for every segment at every point, compiled by Numba and parallelised. Two regularisations are available and are used for different purposes (Sect. 4.6): a hard *cutoff* d , below which a segment induces nothing, and a smooth *vortex core* of radius r_c , which multiplies (4.1) by $h^2/(h^2 + r_c^2)$ with h the distance to the segment (Scully profile).

The boundary condition at control point k with normal $\mathbf{n}_k$ is

$$\sum_l a_{kl} \Gamma_l = -(\mathbf{u}_\infty + \mathbf{v}_{\text{wake}}) \cdot \mathbf{n}_k, \quad (4.2)$$

where $\mathbf{a}_{kl}$ is the normal velocity induced at $\mathbf{k}$ by ring $\mathbf{l}$ with unit circulation. The matrix $\mathbf{A} = [\mathbf{a}_{kl}]$ depends only on the geometry and is inverted once. The right-hand side contains the free stream and the velocity induced by the wake rings whose strengths are already known.

4.4 Wake and time stepping (`VLMSolver._time_sim`)

The wing is started impulsively and marched in time with a step Δt . At each step:

1. the existing wake is convected downstream by $\mathbf{u}_\infty \Delta t$;
2. one new row of rings is shed from the trailing edge, and one new column from each tip if tip shedding is enabled;
3. the circulations are solved from (4.2). The rings shed at *this* step carry the *current* circulation of the panels they are shed from, and their influence is placed in $\mathbf{A}$ rather than in the right-hand side. This is the implicit Kutta condition of Katz and Plotkin (Sect. 13.12). It matters: with an explicit treatment the tip vorticity and the shed vorticity cancel only with a one-step lag, and the tip circulation then relaxes at a rate proportional to the tip panel width, so that a fine mesh never reaches steady state within a reasonable travel (Sect. 4.7);
4. the wake corner points are convected by the velocity induced by the whole system, with the vortex core of radius r_c : this is the roll-up of the wake into tip vortices.

Tip shedding. Ordinary vortex lattice methods shed vorticity from the trailing edge only. For a low-aspect-ratio wing, or a strut with a free tip, the flow also separates along the side edge. The code sheds a column of rings from each tip whose strength equals the circulation of the adjacent panels, exactly as at the trailing edge. The wing's tip-edge vorticity is then cancelled by the shed edge, and the tip vortex leaves along the whole side edge instead of at the trailing corner. This raises the lift of a wing of aspect ratio one at 10° from 0.26 to 0.39 and makes the sectional lift finite at the tip; both are the expected effect of side-edge separation, and the value agrees with the vortex lattice method of Rodriguez (1990) that has the same feature.

4.5 Free surface

The free surface is represented by an image of the whole lattice and wake in the plane $z = 0$. Two images are available. The *antisymmetric* image, in which the image rings have the same circulation as the real ones, makes the pressure on the plane atmospheric; this is the exact condition in the limit of high Froude number, where gravity is too weak to deform the surface on the time scale of the flow. The *symmetric* image, with opposite circulation, makes the plane a rigid wall, the low-Froude limit. The internship report validated the antisymmetric image against measured lift for $Fn_c \geq 1.5$ and found neither image adequate at $Fn_c \approx 1$. The code warns when a case is run below $Fn_c = 1.5$. A Froude-dependent free surface was considered and deferred, because the intended applications lie above that limit and a free-surface panel method would multiply the cost by two to four.

For the inception and cavity models the free surface enters a second time, through the hydrostatic pressure: a cavity open to the atmosphere at depth z has the cavitation number

$$\sigma_c(z) = \frac{p_{\text{atm}} + \rho g|z| - p_{\text{atm}}}{\frac{1}{2} \rho u_\infty^2} = \frac{2g|z|}{u_\infty^2} = \frac{2}{Fn_h^2} \frac{|z|}{h}. \quad (4.3)$$

It is zero at the waterline and largest at the tip.

4.6 Loads and numerical parameters (`VLMSolver._kuttas_loads`)

The load on each bound and trailing segment follows from the linearised Kutta–Joukowski theorem, $\mathbf{F} = \rho \mathbf{u}_\infty \times \mathbf{\Gamma}$ (Katz and Plotkin, Eq. 12.25). The velocity induced by the lattice itself at a segment midpoint is not used for the lift: it is of order $\mathbf{\Gamma}/b$ for a panel of width b and does not converge under refinement. It is kept in a separate vector, `surface.loads_induced`, which carries the induced drag. The sectional lift coefficient of each strip, `surface.Cl_2d`, is the lift of its bound segments on the strip area.

Two numerical parameters are passed to the solver.

- **a_ratio** (default 0.4): the cutoff d of the boundary condition, in units of the smallest panel dimension $b_{\min}$ (width or chord). The influence matrix and the wake term of the right-hand side must use the *same* cutoff, because the wing's tip-edge segment sits in the matrix and the coincident shed edge in the right-hand side, and they must cancel. It must stay below 0.5, or a control point falls inside the cutoff of its own ring.
- **ratio** (default 0.05): the vortex core radius r_c for the wake roll-up and the induced loads, in units of the mean chord. A physical length rather than a mesh length, so the results converge as the mesh is refined. The lift varies by about 1 % for $r_c/\bar{c}$ between 0.02 and 0.1.

Recommended settings: cosine spacing, $u_\infty \Delta t / c \leq 0.05$, at least three chords of travel for a steady case.

4.7 Verification of the wetted solver

The revision of the solver is documented in `docs/tip_loading/report.html`, with the study that produced Table 4.1. The test is the wing of aspect ratio one at 10° with both tips shedding.

Table 4.1. Tip sectional lift and lift coefficient of the revised solver.

Variation	Tip C_l	C_L
Core radius $r_c/c = 0.02, 0.05, 0.10$	0.393, 0.386, 0.369	0.387, 0.386, 0.381
$N = 16, 30, 50, 80$ spanwise panels	0.382, 0.386, 0.388, 0.390	0.386, 0.386, 0.387, 0.388
$u_\infty \Delta t / c = 0.1, 0.05, 0.025$	0.386, 0.374, 0.375	0.386, 0.382, 0.383
Reference: Rodriguez (1990) VLM; Bartlett and Vidal (1955) experiment		0.37; 0.33

Before the revision the tip value ranged from -0.21 to 0.49 over the same variations. The lift coefficient is 4 % above Rodriguez and 17 % above the measurement, the latter difference being attributed in the report to flow separation the potential-flow model cannot capture.

5 Section data

5.1 Why they are needed

The lattice is inviscid. It cannot tell whether the boundary layer has separated, and its pressure at the leading edge is singular. Ventilation inception depends on both. The code therefore reads, for the section in use, three functions of incidence and Reynolds number:

- $\alpha_{\text{sep}}(Re_c)$, the incidence at which the suction side separates over a large part of the chord, taken as the incidence of maximum lift (stall);
- $C_{p,\min}(\alpha, Re_c)$, the minimum pressure coefficient on the suction side with the boundary layer present;
- $x_{\text{sep}}(\alpha, Re_c)$, the chordwise position of turbulent separation.

5.2 NeuralFoil

XFOIL (Drela, 1989) is the standard tool for these quantities, but it is a Fortran program that must be compiled. NeuralFoil (Sharpe, 2025) is a neural network trained on about eight million XFOIL runs; it is pure Python, installs

with pip, and returns in a few milliseconds the force coefficients, the transition points, the boundary-layer edge velocity u_e and shape factor H at 32 stations on each surface, and a *confidence* between 0 and 1 that falls where XFOIL itself would not have converged.

`SectionData.from_neuralfoil(coordinates, name)` builds the table for any section from its coordinates: $C_{p,\min} = 1 - \max(u_e/u_\infty)^2$ on the suction side, α_{sep} from the first lift maximum, x_{sep} from the first station downstream of transition where $H > 2.8$. The table is a JSON file with the grids and the arrays, so that XFOIL polars can be written in the same format when a compiled XFOIL is available (the conda-forge package `libxfoil` or the PyPI package `xfoil` on a machine with gfortran). `SectionData.naca0009()` loads the NACA 0009 table supplied with the code.

Table 5.1. NACA 0009 from NeuralFoil against published values.

Re_c	α_{sep} , NeuralFoil	published	confidence at stall	$C_{p,\min}$ at 10°
10^5	8.5°	9° (Sheldahl and Klimas, 1981)	0.62	-1.3
5×10^5	11.0°	10 to 11°	0.84	-4.7
10^6	13.0°	12°	0.93	-4.5
1.6×10^6	14.5°	14.5° (Harwood et al., modified section)	0.95	-4.5
3×10^6	16.5°	13 to 14° (Abbott and von Doenhoff, 1959)	0.97	-4.5

The surrogate is within a degree of the published stall up to $Re_c = 1.6 \times 10^6$, which covers the towing-tank experiments, and overestimates it above 3×10^6 as XFOIL does for thin sections. Its confidence is above 0.84 for $Re_c \geq 5 \times 10^5$. The inviscid suction peak at 10° is -9.2 , twice the viscous one; an earlier version of the table used it and was labelled provisional; it is kept as `SectionData.naca0009("provisional")`.

6 Inception model (VLMInception)

6.1 The two necessary conditions

Natural ventilation requires two things at the same time (Harwood et al., 2016; Young et al., 2017): **(A)** a region of separated flow at sub-atmospheric pressure on the suction side, and **(B)** a path by which air can travel from the free surface into that region. Condition A is a property of each section. Condition B depends on how the low-pressure region connects to the surface, and three routes are recognised: *nose* ventilation, when leading-edge separation reaches the waterline; *tail* ventilation, when the depression behind a blunt trailing edge is amplified into a trough that reaches the separated flow; *tip-vortex* ventilation, when the tip vortex of a submerged element is strong enough and shallow enough to reach the surface.

6.2 Condition A on each section

After a wetted run, the model computes for each spanwise section j its depth z_j , its cavitation number from (4.3), its Reynolds number, and its effective incidence from the sectional lift of the lattice, $\alpha_{\text{eff},j} = C_{l,j}/(2\pi \cos \lambda)$. The section is **separated** if $\alpha_{\text{gate}} \geq \alpha_{\text{sep}}(Re_c)$ and **sub-atmospheric** if $-C_{p,\min}(\alpha_{\text{eff}}, Re_c) \geq \sigma_c(z_j)$. It is *ventilation-prone* when both hold.

The incidence α_{gate} that decides separation is a choice, because the two published data sets disagree (Sect. 6.4). With `seal="effective"` it is the section's own effective incidence, which on a strut of immersed aspect ratio one is about

half the geometric one because the free surface unloads the waterline. With `seal="geometric"` it is the geometric incidence, as in Viola's `bem-fem-fsi`.

6.3 Condition B, the routes

- **Nose:** a prone section lies within the waterline band $|z| \leq 0.5 c$. The band is a parameter; the result is insensitive to it between 0.25 and 1.0 chord.
- **Tail:** the trailing edge is blunt (`surface.blunt_te = True`), its base pressure $C_{p,\text{base}} = -0.3$ is sub-atmospheric within the band, and $Fn_h > AR_h^{-1/2}$, the condition under which the surface depression grows (Aguiar Ferreira et al., 2026).
- **Tip vortex** (`submerged=True`): the vortex strength is the largest sectional circulation of the element, the core pressure $C_{p,\text{core}} = -\Gamma^2/(4\pi^2 r_c^2 u_\infty^2)$ is sub-atmospheric at the shallowest point of the tip filament of the free wake, and that point is within one core radius of the surface. No published experiment isolates this route; the code labels it unvalidated.

The call `VLMInception.assess(surface, solver, data, g, nu, band, seal, submerged)` returns the per-section state, the fraction of the span that is prone, the state of each route, the confidence of the section data, and the flags `incepts` and `route`. `VLMInception.report` prints it as a table.

6.4 Validation against published values

Figure 6.1 and Table 6.1 give the predicted inception incidence of the strut of Harwood et al. (NACA 0009-type section, chord 0.279 m, $AR_h = 1$) against Froude number for the two gates. Harwood et al. formed cavities at about 14.5° , independent of Fn_h , during acceleration and with perturbations; Aguiar Ferreira et al. measured spontaneous inception under steady conditions at 20° to above 25° , rising with Froude number. The geometric gate reproduces the first within 1 to 2° , the effective gate the second. The tail threshold is implemented as published and its case only checks the implementation.

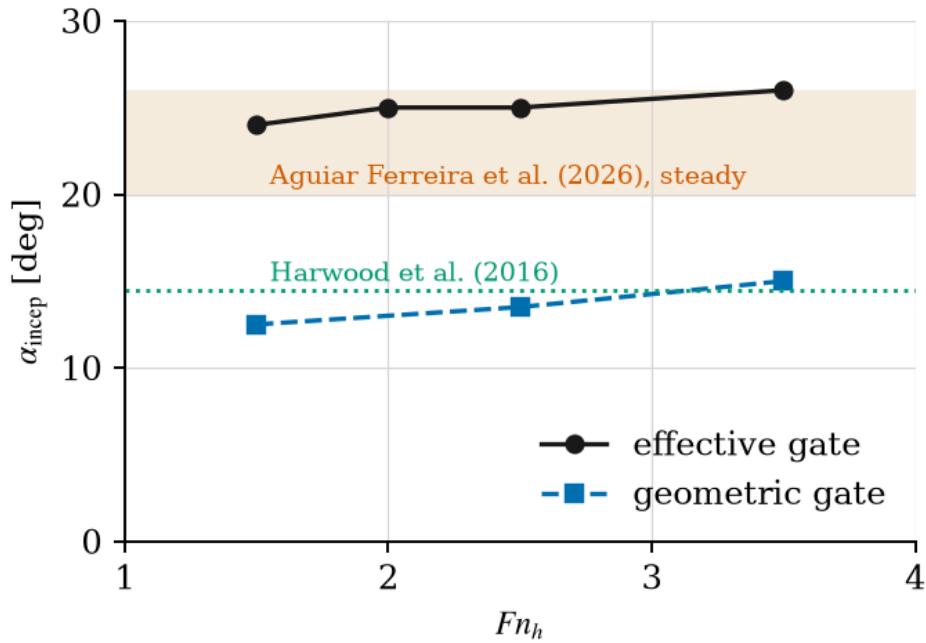


Figure 6.1. Predicted inception incidence versus depth Froude number for the two separation gates, strut of Harwood et al., $AR_h = 1$, with the two published ranges.

Table 6.1. Predicted inception incidence.

Fn_h	Re_c	effective gate	geometric gate
1.5	6.9×10^5	24°	12.5°
2.5	1.2×10^6	25°	13.5°
3.5	1.6×10^6	26°	15.0°

7 Cavity model (`VLMCavity.CavitySolver`)

7.1 A cavity on the lattice

A ventilated cavity is a region of the suction side where the pressure is atmospheric, so $C_p = -\sigma_c(z)$ there, and where the flow leaves the surface and follows the cavity boundary. In linearised thin-section theory (Acosta, 1955; Tulin, 1953) it is represented on the mean surface by a distribution of sources $\mathbf{q}$, which gives the cavity a thickness, on top of the vorticity that represents the lift. The code follows this on the lattice. A ventilated panel carries, besides its ring, a constant-strength source sheet, and two conditions hold on it instead of one:

$$(\mathbf{u}_\infty + \mathbf{v}_{\text{rings}} + \mathbf{v}_{\text{sources}}) \cdot \mathbf{n}_p = 0 \quad \text{on the wetted side (kinematic),} \quad (7.1)$$

$$\mathbf{u}_\infty \cdot \boldsymbol{\tau} + \mathbf{v}_{\text{wake}} \cdot \boldsymbol{\tau} + \mathbf{v}_{\text{sources}} \cdot \boldsymbol{\tau} + \frac{1}{2}\gamma_i = u_\infty \left(1 + \frac{1}{2}\sigma_c\right) \quad \text{on the cavity side (dynamic).} \quad (7.2)$$

Here $\gamma_i = (\Gamma_i - \Gamma_{i-1})/\Delta c_i$ is the bound vorticity per unit chord of the panel, and $\pm\gamma_i/2$ is the jump of tangential velocity across the sheet, which is the only tangential velocity a planar lattice induces on itself. The source sheet emits $\mathbf{q}/2$ on each side; the normal velocity of (7.1) is evaluated at a point displaced to the wetted side, so the own-panel term is included. Equation (7.2) is linearised as in Acosta, so that the verification of Sect. 7.4 compares like with like. The unknowns are all ring strengths plus the source strengths of the cavity panels; the system is square and solved directly at each step.

The velocity of a constant-strength source quadrilateral is that of Hess and Smith (1962). The in-plane part uses the edge logarithms of Katz and Plotkin (Sect. 10.4.1); the normal part is the solid angle of the panel at the point, $\mathbf{w} = \Omega/4\pi$, which the code computes from two triangles and which has no special case for edges parallel to an axis. Its checks are in `tests/test_cavity.py`: $\mathbf{w} = \pm\frac{1}{2}$ on the two sides, the far field of a point source, and the two-dimensional sheet limit.

7.2 Cavity length

The chordwise length L_j of the cavity on section j is taken from the sectional theory. With $\Psi_j = \sigma_c(z_j)/(2\alpha_{\text{eff},j})$, where α_{eff} is that of the wetted solution of the same step, $L_j = L(\Psi_j)$ from the blend of Acosta's partial-cavity and Tulin's supercavity solutions used by Harwood et al. (module `vent_section`, reused verbatim from Viola's `bem-fem-fsi`, MIT licence). The cavity covers the whole panels up to L_j and the load is interpolated between the two bracketing panel counts, so it is continuous in the incidence.

An iteration of the length on the lattice, adjusting it until the cavity thickness vanishes at its end, was implemented and abandoned: the source strengths of a collocated source–vortex lattice alternate in sign from panel to panel, and the thickness at closure is noise. The closure residual is kept as a diagnostic.

7.3 Two branches

The linear closed-cavity theory holds for $L \leq 0.5$. Up to that length the section is solved on the lattice as above, and the lift follows from the redistributed circulation. Beyond it, the sectional lift slope $a_0(L)$ of the blend is imposed on the section by scaling the free-stream term of its kinematic condition by $a_0/2\pi$, which is the effective-incidence correction of the internship report in a conservative form. The two branches agree to 4 % at $L = 0.5$. The length used for the loads is capped at ten chords, where a_0 is within 1 % of its supercavitating limit $\pi/2$.

The centre of pressure e of each section is computed from the chordwise distribution of bound vorticity on the lattice branch, and from the sectional blend, which tends to $3c/16$ for a supercavity, on the sectional branch.

7.4 Verification against Acosta

A rectangular wing of aspect ratio 40 with a uniform prescribed cavitation number approximates a two-dimensional flat plate. Figure 7.1 compares the lift slope of its mid-span section with Acosta's closed form

$a_0 = \pi(1 + 1/\sqrt{1-L})$. The lattice is 4 to 5 % high over the whole range and reproduces the rise of the lift slope above 2π that a short cavity produces by its displacement.

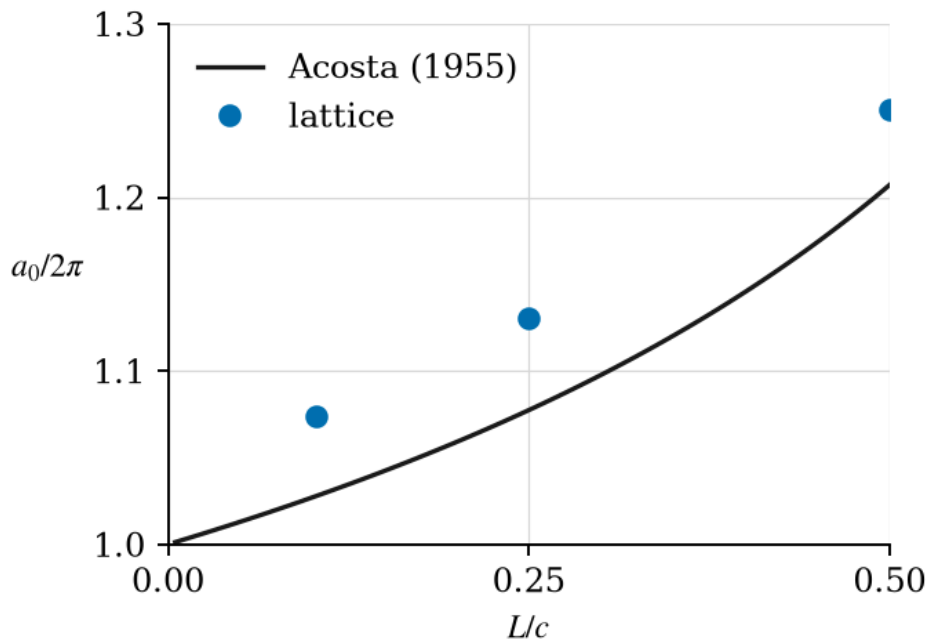


Figure 7.1. Lift slope ratio versus cavity length for a partial cavity from the leading edge, lattice against Acosta (1955); aspect ratio 40, $\alpha = 4^\circ$, 16 chordwise panels.

8 Regime machine (VLMCavity.Ventilation)

8.1 States and rules

The flow on each surface is in one of three regimes: fully wetted (FW), partially ventilated (PV) or fully ventilated (FV). The machine follows Harwood et al. (2016) and the design of `bem-fem-fsi`.

- **Formation.** A cavity forms when the inception signal fires. The signal is any function `(surface, solver) -> assessment` returning the dictionary of `VLMInception.assess`; two simple ones are provided,

`stall_gate(alpha_deg)` and `vent.force(surface)` , the latter being the perturbation route of the experiments. The cavity is seeded on the prone sections that can hold one, and grows at a rate limit (default 0.2 chords of cavity per chord of travel), so that formation takes a few convective times as observed.

- **Persistence.** A ventilated section keeps its cavity as long as a cavity can exist there (sectional length of at least one panel). It never asks again for the seal. This asymmetry is the hysteresis: a cavity survives to incidences well below the one at which it formed.
- **Elimination.** Harwood et al. found that a fully ventilated cavity washes out when the re-entrant jet at its closure has an upstream component, that is when the closure line makes more than 45° with the flow. The code evaluates the local angle of the closure line at the representative mid-depth section from the sectional lengths, using the long-cavity relation $L = 2.31/\Psi$ that Harwood et al. used to derive their washout boundary. Above the formation boundary an unstable cavity persists and sheds (PV); below it the flow rewets after the cavity has been unstable for three chords of travel.
- **Label.** FW when nothing is ventilated; FV when the cavity reaches the tip and the closure line is stable; PV otherwise.

A sweep in incidence or speed carries the state from one run to the next with `carry_state` .

8.2 Validation on the strut of Harwood et al.

Table 8.1 gives the fully ventilated loads at $Fn_h = 2.5$ against the semi-empirical relation of Damley-Strnad, Harwood and Young (2019), which was fitted to the towing-tank data of several campaigns, and Figure 8.1 the hysteresis loop.

Table 8.1. Ventilated to wetted lift ratio, $AR_h = 1$, $Fn_h = 2.5$.

α	C_L wetted	C_L ventilated	ratio, this code	ratio, Damley-Strnad et al.	e wetted	e ventilated
6°	0.162	0.133	0.82	0.96	0.224	0.191
10°	0.299	0.190	0.64	0.69	0.203	0.188
14°	0.463	0.244	0.53	0.57	0.189	0.188
18°	0.661	0.296	0.45	0.53	0.174	0.188

The code is 5 to 15 % below the semi-empirical relation with the same trend; both lie in the 50 to 70 % loss reported at high incidence. Against the measured lift curves at the same condition, digitised later, the code is within 1 % on average (Sect. 10.4). At the published washout condition, $\alpha = 20^\circ$ and $Fn_h = 1.5$, the computed closure angle at mid-depth is 34.5° against the measured 40.75° .

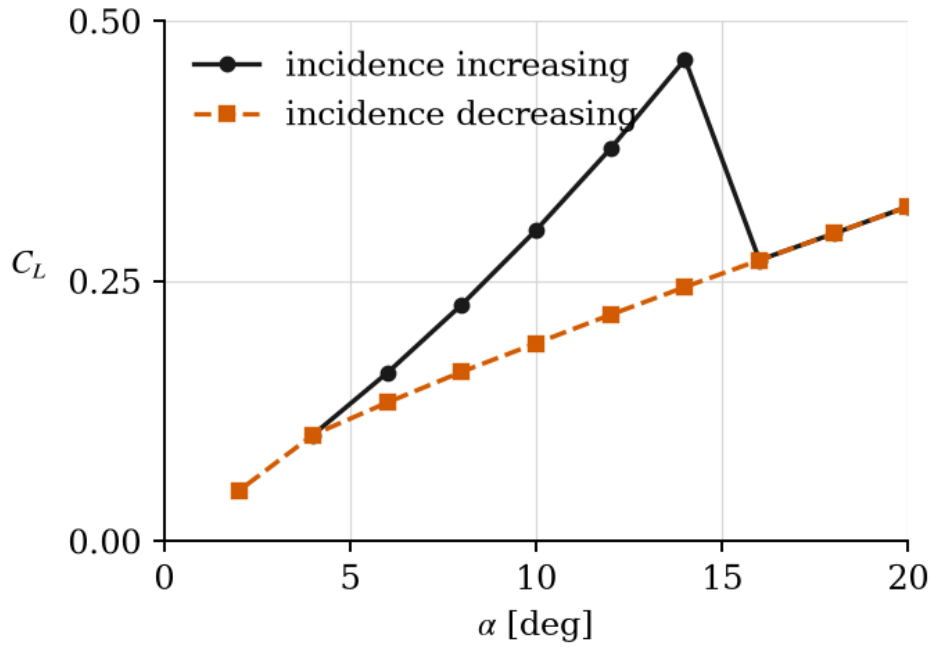


Figure 8.1. Lift coefficient versus incidence swept up from a wetted state and down from a ventilated one, strut of Harwood et al., $Fn_h = 2.5$, stall gate at 14.5° .

The ascending branch stays wetted to 14° and ventilates at 16° ; the descending branch stays fully ventilated to 10° , sheds partially at 8° and 6° , and rewets at 4° . The bistable band, 6° to 14° , is the one Harwood et al. report at this Froude number, and `bem-fem-fsi` computes 4° to 14° .

9 A complete run

`docs/report/example_end_to_end.py` runs the three layers together on the strut of Harwood et al. at $Fn_h = 2.5$: the inception model with the geometric gate supplies the signal, the regime machine forms and eliminates the cavity, the cavity model gives the loads. The essential lines are:

```
data = SectionData.naca0009() # section data (Sect. 5)

def inception(surface, solver): # the signal (Sect. 6)
    return vi.assess(surface, solver, data, g=9.81, nu=1e-6, band=0.5, seal="geometric")

strut = VLMSurface(np.zeros(3), 0, True, "both", 2*h, np.ones(N + 1)*c, 0.0, np.deg2rad(alpha),
    0, 0, 0, False, True, N, M); strut._build_wing()
vlm = CavitySolver([strut], np.array([u, 0, 0]), "antisymmetric", 0.05, 0.4, g=9.81,
    rate=3.0, inception=inception) # cavity + regime (Sects. 7, 8)
carry_state(previous_vlm, previous_strut, vlm, strut) # hysteresis across the sweep
vlm._time_sim(4*c/u, 0.05*c/u, "classic"); vlm._kuttas_loads()
vlm.vent.report(strut)
```

The output for the sweep is in Table 9.1 (`example_end_to_end.json`).

Table 9.1. End-to-end sweep, strut of Harwood et al., $Fn_h = 2.5$.

sweep	α	C_L	regime	route	ventilated fraction	τ at mid-depth	ϵ at mid-depth
up	4°	0.102	FW	—	0.00	0.00	0.290

sweep	α	C_L	regime	route	ventilated fraction	τ at mid-depth	ϵ at mid-depth
up	6°	0.162	FW	–	0.00	0.00	0.285
up	8°	0.228	FW	–	0.00	0.00	0.278
up	10°	0.299	FW	–	0.00	0.00	0.272
up	12°	0.378	FW	–	0.00	0.00	0.265
up	14°	0.245	FV	nose	1.00	1.50	0.188
up	16°	0.270	FV	nose	1.00	1.64	0.188
up	18°	0.296	FV	nose	1.00	1.81	0.188
up	20°	0.322	FV	nose	1.00	2.00	0.188
down	18°	0.296	FV	nose	1.00	1.81	0.188
down	16°	0.270	FV	nose	1.00	1.64	0.188
down	14°	0.245	FV	nose	1.00	1.50	0.188
down	12°	0.218	FV	nose	1.00	1.37	0.188
down	10°	0.190	FV	nose	1.00	1.26	0.188
down	8°	0.163	PV	nose	0.92	1.17	0.188
down	6°	0.133	PV	nose	0.83	1.07	0.188
down	4°	0.102	FW	washout	0.00	0.00	0.291
down	2°	0.048	FW	washout	0.00	0.00	0.296

With the inception model as the signal, the cavity forms at 14° by the nose route (the section data give a stall incidence of 13.5° at this Reynolds number), one step earlier than with the 14.5° stall gate of Sect. 8.2; the descending branch is unchanged, with washout at 4°. The centre of pressure moves from 0.27 to 0.29 chords forward of mid-chord wetted to 0.19 ventilated, the supercavitating value **3c/16**.

10 Validation against measured curves

10.1 What is tested, and by which data

The checks of Sects. 4 to 8 compared the code with *single values* quoted in the papers: a lift coefficient at one incidence, a closure angle at one condition. A validation needs the measured *curves*, so that trends with incidence, Froude number and aspect ratio are tested and not only one point on them. Table 10.1 lists the curves that would test each part of the code, in order of value. Orain (2026) digitised items 1, 5, 6 and 9 from the published figures; Sects. 10.2 to 10.5 compare them with the code and Sect. 10.6 draws the conclusions. The remaining items are still to be digitised, and Sect. 10.7 explains how.

Table 10.1. Measured curves that test the code, and the state of each comparison.

#	Source	Quantity	Conditions	What it tests	Script	State
1	Harwood et al. (2016), fully wetted lift curves	$C_L(\alpha)$, FW	$AR_h = 0.5, 1, 1.5$; $Fn_h = 1.0$ to 4.5	the wetted solver with the free-surface image, on which everything else rests	docs/validation/validate.py, Case B	done, Sect. 10.3

#	Source	Quantity	Conditions	What it tests	Script	State
2	Harwood et al. (2016), Fig. 16, regime map	regime of each point and how it was reached	α versus Fn_h , $AR_h = 1$	the geometric gate and the bistable band (Sects. 6, 8)	docs/inception/validate.py , Case 1c	to digitise
3	Aguiar Ferreira et al. (2026), inception angle against Froude number, stability map	$\alpha_{incept}(Fr)$ with the trigger of each point	two sections, $AR = 1, 1.5$	the effective gate, the Reynolds trend, the tail threshold	docs/inception/validate.py , Cases 1, 2	to digitise
4	Damley-Strnad et al. (2019), Fig. 5	inception and rewetting C_L against Fn_h	compiled campaigns	a global check over a wide Froude range	new case in docs/inception/validate.py	to digitise
5	Harwood et al. (2016), fully ventilated lift; drag and yawing moment; Damley-Strnad et al. (2019), Fig. 3	$C_L, C_D, C_M(\alpha)$, FV	$Fn_h = 1.0$ to 4.5 ; $AR_h = 0.5, 1, 1.5$	the cavity loads and, through the moment, the centre of pressure	docs/validation/validate.py , Case C	lift done, Sect. 10.4; drag and moment to digitise
6	Harwood et al. (2016), cavity photograph, closure angles of their Fig. 8	$L(z)$; local closure angles	$\alpha = 10^\circ$, $Fn_h = 1.5$, $AR_h = 1$	the only local check of the cavity model (Sect. 7.2)	docs/validation/validate.py , Case D	profile done, Sect. 10.5; closure angles to digitise
7	Harwood et al. (2016), washout points and one hysteresis loop	Fn_h at washout against C_L ; $C_L(\alpha)$ up and down	fixed Fn_h	the re-entrant-jet criterion and the band width (Sect. 8)	docs/cavity/validate.py , Cases B, C	to digitise
8	Wadlin, Ramsen and Vaughan (1955); Kiceniuk (1954)	lift against depth of submersion; onset of ventilation	plates and hydrofoils beneath the surface	the tip-vortex route, which has no validation at all	new case in docs/inception/validate.py	to digitise
9	Rodriguez (1990); Lamar (1974); Bertin and Smith (1998); Weber and Brebner (1958)	$C_L(\alpha)$ of wings without free surface	rectangular $AR = 1, 3$; 45° swept $AR = 5$	the wetted lattice, tip shedding and sweep, without the free surface	docs/validation/validate.py , Case A	done, Sect. 10.2

The digitised curves are in `docs/validation/data/`, the raw files with their provenance in `docs/validation/data/raw/`, and `docs/validation/validate.py` runs the code at the conditions of every point and writes `validate.json` and Figures 10.1 to 10.4 (25 min). All runs use the settings of the report: on the strut, 12 sections over the immersion and eight chordwise panels, cosine spacing, a time step of 0.05 chords and three chords of travel. Two measures are used throughout. The *lift slope* is the slope of a straight line through the origin fitted to the lift curve up to 10° , in units of lift coefficient per radian of incidence; a flat plate of infinite span has $2\pi = 6.28$. The *mean difference* is the average, over the incidences of the reference curve, of the computed lift minus the reference lift, divided by the reference lift; a positive value means the code predicts more lift than the reference.

10.2 Wings without a free surface

The first test removes the free surface, the section data and the cavity, and asks whether the lattice alone gives the lift of a wing. Three wings are run: rectangular wings of aspect ratio 1 and 3, for which Rodriguez (1990) published a free-wake vortex lattice with tip shedding and Lamar (1974) reported wind-tunnel measurements, and a rectangular wing of aspect ratio 5 swept back by 45° , the textbook example of Bertin and Smith (1998), for which a straight-wake vortex lattice result and the measurements of Weber and Brebner (1958) exist. Each wing has 15 spanwise panels per

half-span and five chordwise panels and travels at least three chords and two spans; doubling the travel changes the lift at 8° by 0.5 % or less, so the runs have reached steady state.

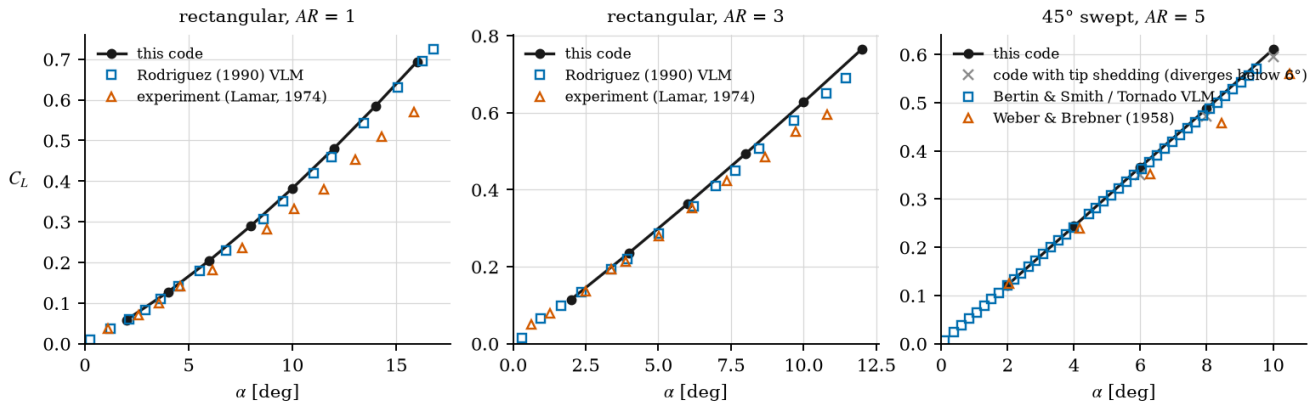


Figure 10.1. Lift coefficient versus incidence for the code, the reference vortex lattice methods and the experiments: (a) rectangular wing, $AR = 1$; (b) rectangular wing, $AR = 3$; (c) 45° swept wing, $AR = 5$. In (c) the crosses are the code with tip shedding, which diverges below 6° .

Table 10.2. Lift slope per radian of the code and of the references, and mean difference of the code from each reference.

Wing	Code	Vortex lattice reference	Experiment	Code minus lattice	Code minus experiment
rectangular, $AR = 1$, tip shedding	2.08	1.97, Rodriguez (1990), free wake	1.77, Lamar (1974)	+3 %	+14 %
rectangular, $AR = 3$, tip shedding	3.54	3.37, Rodriguez (1990), free wake	3.24, Lamar (1974)	+4 %	+7 %
45° swept, $AR = 5$, no tip shedding	3.50	3.45, Bertin and Smith (1998), Tornado	3.17, Weber and Brebner (1958)	+1 %	+10 %

On the two rectangular wings the code lies 3 to 4 % above the free-wake lattice of Rodriguez over the whole range of incidence (Figure 10.1a, b; Table 10.2), and reproduces the curved lift curve of the wing of aspect ratio one: its slope rises from 1.7 per radian at 2° to 2.5 per radian at 16° , because the vortices shed from the side edges add lift as the incidence grows. The excess over the experiments, 14 % at $AR = 1$ and 7 % at $AR = 3$, is the difference between a potential-flow lattice and a real wing with a boundary layer, and is of the same size as the 17 % noted in Sect. 4.7 for the measurement of Bartlett and Vidal (1955). On the swept wing without tip shedding the code is within 1 % of the straight-wake lattice and 10 % above the wind-tunnel measurement (Figure 10.1c). The sweep is therefore implemented correctly.

Tip shedding must be off on a swept wing. With tip shedding and cosine spacing, the swept wing diverges at 2° and 4° within ten time steps, giving lift coefficients of -879 and -498 after ten chords of travel, while it converges at 6° and above and is then within 3 % of the straight-wake lattice (crosses in Figure 10.1c). The mechanism is the following. Tip shedding relies on the cancellation, at the tip control point, between the wing's tip-edge segment, which is in the influence matrix, and the shed column that coincides with it, which is in the right-hand side (Sect. 4.6). This cancellation holds as long as the shed column stays on the line of the edge to within a fraction of the cutoff distance, which is 0.4 of the tip panel width, or 0.0015 spans with cosine spacing. On the swept wing the tip panel carries 2.7 times the circulation of its neighbour after the first step, against 0.65 times on the unswept wing of the same aspect ratio, and the roll-up moves the first shed corners by 0.0013 spans, of the order of the cutoff; the cancellation is lost and the tip circulation grows without bound. The instability does not occur with uniform spacing, without sweep, or without tip shedding. Tip shedding models the separation at the side edge of a low-aspect-ratio wing or a strut; on a

swept wing of moderate aspect ratio, where the flow is attached at the tip, it is neither needed nor stable, and `shedding="none"` should be used.

10.3 The wetted strut

The second test adds the free surface. The strut of Harwood et al. (2016), of chord 0.2794 m, is run fully wetted at immersed aspect ratios 0.5, 1 and 1.5 and incidences from 2.5° to 15°, with the lift coefficient on the immersed area $h c$, as in the paper. The antisymmetric image of Sect. 4.5 does not depend on the Froude number, so one computed curve per aspect ratio is compared with the measurements at every Froude number (Figure 10.2, Table 10.3). Harwood et al. also published the lift of their own lifting-line model, which uses the same image; it is shown for $AR_h = 1$.

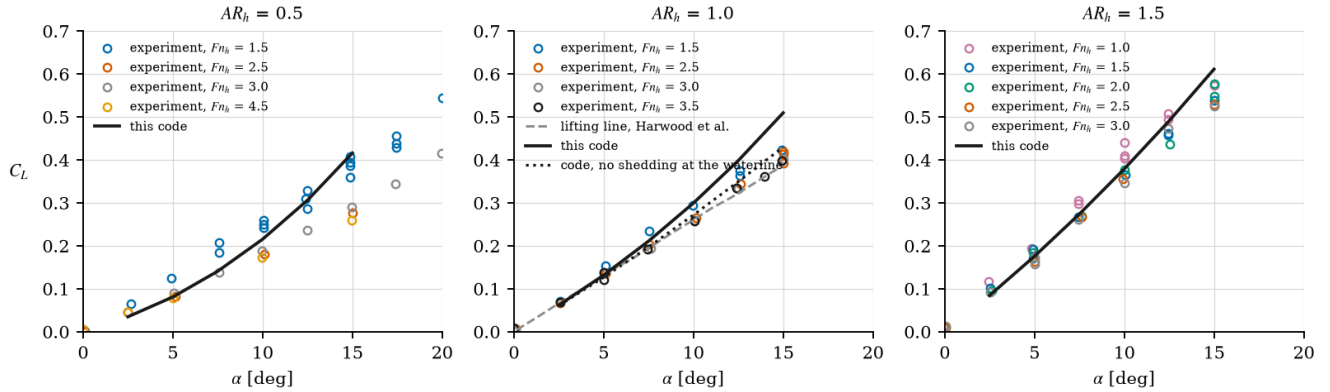


Figure 10.2. Lift coefficient versus incidence of the fully wetted strut, code against the measurements of Harwood et al. (2016) at every Froude number and their lifting-line model: (a) $AR_h = 0.5$; (b) $AR_h = 1$; (c) $AR_h = 1.5$. The dotted line in (b) is the code without vortex shedding from the waterline edge.

Table 10.3. Lift slope per radian of the code and, for each Froude number, of the measurements of Harwood et al. (2016), with the mean difference of the code from the measurements in brackets.

AR_h	Code	$Fn_h = 1.0$	1.5	2.0	2.5	3.0	3.5	4.5
0.5	1.13	–	1.48 (–21 %)	–	0.93 (+16 %)	1.06 (+9 %)	–	0.98 (+11 %)
1.0	1.65	–	1.69 (–2 %)	–	1.53 (+9 %)	1.48 (+10 %)	1.48 (+9 %)	–
1.5	2.12	2.37 (–10 %)	2.09 (0 %)	2.11 (+1 %)	2.01 (+7 %)	1.97 (+5 %)	–	–

The measured lift slope falls as the Froude number rises, by 13 % between $Fn_h = 1.5$ and 3.5 at $AR_h = 1$, and settles above $Fn_h = 2.5$. This is the Froude effect that the image cannot represent: at low Froude number the surface rises on the pressure side and falls on the suction side, which adds lift. The code matches the measurements at $Fn_h = 1.5$ within 2 % at $AR_h = 1$ and 1.5, and is 5 to 10 % above them at $Fn_h \geq 2.5$, where the image is meant to apply. The lifting line of Harwood et al., with the same image, gives 1.49 per radian at $AR_h = 1$, the value the measurements settle at; the code is 12 % above it. At $AR_h = 0.5$ the measurements themselves scatter by 50 % between Froude numbers and the code lies within that scatter. A further difference is in the shape of the curves: the computed lift curves are convex, with a slope that rises from 1.40 per radian at 2.5° to 1.94 per radian at 15° for $AR_h = 1$, whereas the measured curves at $Fn_h \geq 2.5$ are straight to within 3 %.

Where the excess comes from. Four variants of the run at $AR_h = 1$ locate the difference (Table 10.4). Doubling the number of sections does not change the slope, so the excess is not a resolution effect. The symmetric image, which makes the free surface a rigid wall, nearly doubles the lift and confirms the choice of Sect. 4.5. Without any tip shedding the slope falls to 1.31, 12 % below the measurements. Shedding from the tip only, and not from the waterline edge, gives 1.52, within 3 % of the high-Froude measurements and of the lifting line; the sectional lift then goes to

zero at the waterline (0.03 at $z/h = 0.01$, against 0.23 with waterline shedding). This is what the physics requires. At high Froude number the free surface is a surface of constant pressure; a section at the waterline has atmospheric pressure on both of its sides, so its lift, which is a pressure difference, must vanish there. Shedding a vortex column from the waterline edge treats that edge like the separated side edge of a wing and keeps a finite loading on it; this is the 8 % excess and part of the convexity. All the results of Sects. 6 to 9 were obtained with shedding from both edges of the strut. They should be rerun with `shedding="right"`, which sheds from the tip only. The lower wetted lift will lower the effective incidence of every section, and with it the predicted inception incidence and the cavity length, by amounts not yet computed.

Table 10.4. Numerical variants of the wetted strut at $AR_h = 1$: lift slope per radian and lift coefficient at 10° .

Variant	Slope	C_L at 10°
settings of the report: shedding from both edges, antisymmetric image, 12 sections	1.65	0.299
24 sections	1.65	0.302
shedding from the tip only	1.52	0.272
no tip shedding	1.31	0.229
symmetric image	2.92	0.522
measurements at $Fr_h = 2.5$ to 3.5	1.48 to 1.53	0.256 to 0.262

10.4 The ventilated strut

The third test is the cavity model. The cavity is imposed on every section, as an experimenter does by perturbing the flow, and the strut is run at incidences from 5° to 30° at each of the 14 combinations of aspect ratio and Froude number measured by Harwood et al. (2016) in the fully ventilated regime (Figure 10.3, Table 10.5).

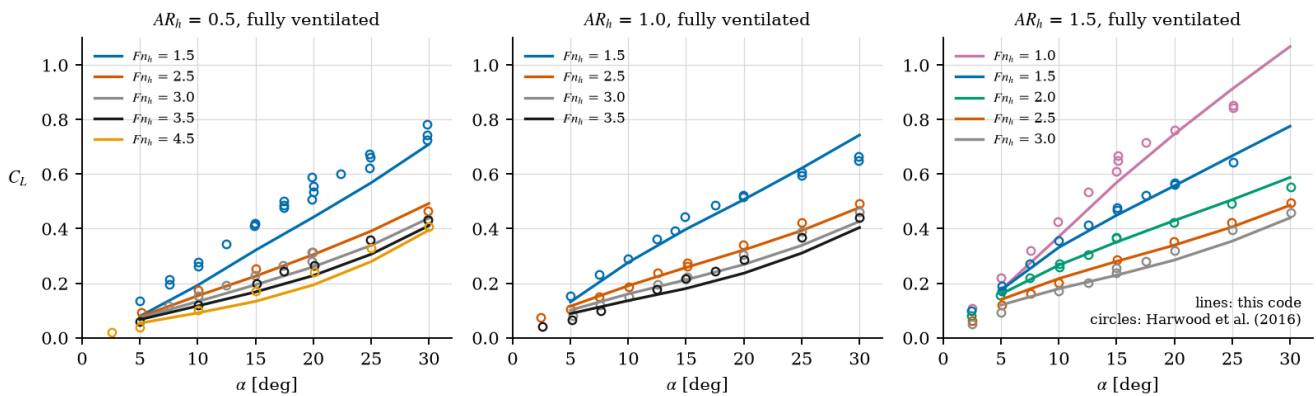


Figure 10.3. Lift coefficient versus incidence of the fully ventilated strut for Froude numbers from 1.0 to 4.5, code (lines) against the measurements of Harwood et al. (2016) (circles): (a) $AR_h = 0.5$; (b) $AR_h = 1$; (c) $AR_h = 1.5$.

Table 10.5. Fully ventilated strut: mean difference and largest absolute difference of the computed lift coefficient from the measurements over the measured range of incidence, and the regime reported by the code at $5^\circ, 10^\circ, \dots, 30^\circ$ (P partially, V fully ventilated).

AR_h	Fr_h	Mean difference	Largest difference in C_L	Regimes
0.5	1.5	-25 %	0.106	PPPPVV
0.5	2.5	-6 %	0.027	PPVVVV
0.5	3.0	-14 %	0.054	PVVVVV
0.5	3.5	-7 %	0.053	PVVVVV

AR_h	Fn_h	Mean difference	Largest difference in C_l	Regimes
0.5	4.5	-5 %	0.049	VVVVVV
1.0	1.5	-4 %	0.047	PPPPVV
1.0	2.5	-1 %	0.029	PVVVVV
1.0	3.0	-6 %	0.050	PVVVVV
1.0	3.5	-3 %	0.059	PVVVVV
1.5	1.0	-7 %	0.069	PPPPPP
1.5	1.5	-1 %	0.026	PPPVVV
1.5	2.0	+2 %	0.037	PPVVVV
1.5	2.5	+2 %	0.022	PVVVVV
1.5	3.0	-1 %	0.041	PVVVVV

At $AR_h = 1$ and 1.5 the ventilated lift is within 7 % of the measurements on average for every Froude number, and the largest single difference, 0.07 in lift coefficient, occurs at $Fn_h = 1.0$, below the Froude number at which the image is valid. The two trends that matter for a foiling boat are reproduced: the lift rises nearly linearly with incidence up to 30° , and falls with the Froude number, by a factor 1.8 in the measurements and 2.1 in the code between $Fn_h = 1.5$ and 3.5 at 20° . Sect. 8.2 found the code 5 to 15 % below the semi-empirical ratio of Damley-Strnad et al. (2019) at $Fn_h = 2.5$; against the measurements at the same condition it is within 1 %. At $AR_h = 0.5$ the code is 5 to 14 % low for $Fn_h \geq 2.5$ and 25 % low at $Fn_h = 1.5$, where the measured curve is also steeper than at the other Froude numbers. The regime reported by the code agrees with the experiment at all points but the lowest: at 5° , and at 10° for $Fn_h \leq 1.5$, the computed cavity does not reach the tip and the code reports the partially ventilated regime where the experiment was fully ventilated; the reason appears in the cavity profile below. The wetted lift of the same strut is 5 to 10 % high at $Fn_h \geq 2.5$ (Sect. 10.3); with the waterline shedding off, the ventilated lift would fall too, by an amount not yet computed.

10.5 The cavity profile

The lift curves test the cavity model as a whole. The photograph of the cavity in Harwood et al. (2016) at $\alpha = 10^\circ$, $Fn_h = 1.5$ and $AR_h = 1$ tests it locally, section by section: Orain (2026) digitised the cavity length against depth from the photograph and from the lifting-line model of the same paper, together with the sectional lift, effective incidence and lift slope of that model. The code is run at the same condition with 12 and 24 sections; the two meshes agree within 0.05 chords at every depth (Figure 10.4).

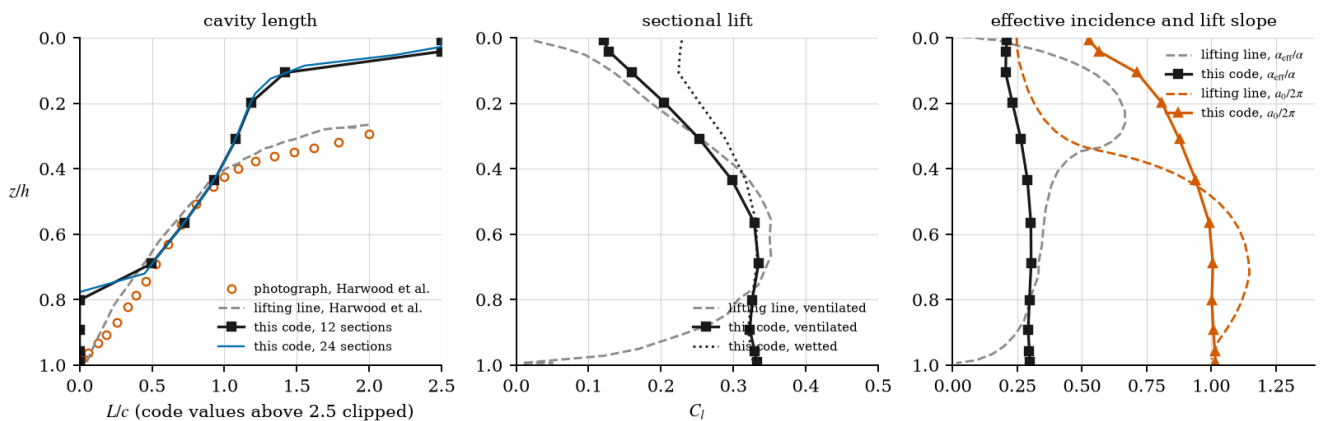


Figure 10.4. Sectional quantities versus depth at $\alpha = 10^\circ$, $Fn_h = 1.5$, $AR_h = 1$: (a) cavity length, code against the photograph and the lifting-line model of Harwood et al. (2016); (b) sectional lift, ventilated and wetted; (c) effective incidence and lift slope, code against the lifting-line model. In the code α_{eff} is $C_l/2\pi$ of the wetted solution and α_0 is the ventilated C_l divided by that α_{eff} .

Between 0.43 and 0.72 of the immersion the computed cavity length is within 10 % of the photograph (0.93 against 0.98 chords at $z/h = 0.43$; 0.50 against 0.53 at 0.69) and closer to it than the lifting-line model, which is 10 to 20 % short over the same range (Figure 10.4a). Two disagreements bound this range. Near the surface, above $z/h = 0.35$, the measured cavity lengthens to two chords, the edge of the photograph, while the computed one is 1.1 chords at $z/h = 0.31$ and exceeds two chords only within 0.05 of the surface; the lifting-line model shows the same defect to a lesser degree. Near the tip, below $z/h = 0.75$, the code has no cavity at all, whereas the photograph shows one that shortens smoothly to zero at the tip (0.37 chords at $z/h = 0.80$, 0.07 at 0.96). The cause is a rule of the regime machine (Sect. 8.1): a section whose sectional cavity length is shorter than one chordwise panel, 0.125 chords here, is declared unable to hold a cavity and is rewetted. The rule cuts the cavity where the measured one still covers a third of the chord, and is why the code reports this case, and the low-incidence points of Sect. 10.4, as partially rather than fully ventilated. The criterion should be on the sectional length itself, not on the panel count. Along the whole depth the cavitation number of the code matches the lifting-line model, as it must, both being $2gz/u_\infty^2$.

The sectional lift of the lifting-line model goes to zero at the surface and at the tip (Figure 10.4b). The code keeps 0.12 at the surface, because of the waterline shedding discussed in Sect. 10.3, and 0.33 at the tip, because of the tip shedding, which the report keeps as the model of side-edge separation on a strut. The ventilated lift slope of the code falls to half of 2π at the surface and not to the supercavitating quarter that the lifting-line model reaches (Figure 10.4c), because on the lattice the circulation that a section loses is partly taken up by its neighbours.

10.6 What the validation shows

1. **The lattice is right.** On wings without a free surface it is within 4 % of two published vortex lattice methods and 7 to 14 % above wind-tunnel measurements, the usual excess of potential flow. Sweep is correct. Tip shedding is for low-aspect-ratio unswept wings and struts only: on a swept wing with cosine spacing it diverges below 6° .
2. **The free-surface image is right at high Froude number, once the waterline edge is treated correctly.** With shedding from the tip only, the wetted strut is within 3 % of the measurements at $Fn_h \geq 2.5$ and of Harwood's lifting line, and the loading vanishes at the surface as it should. With shedding from both edges, the setting used so far, it is 5 to 10 % high. The measured 13 % fall of lift slope between $Fn_h = 1.5$ and 3.5 is a Froude effect the image cannot represent.
3. **The cavity model gives the ventilated lift.** Within 7 % of the measurements at $AR_h = 1$ and 1.5 over 14 combinations of Froude number and aspect ratio and incidences from 5° to 30° ; 5 to 25 % low at $AR_h = 0.5$.
4. **The cavity shape is right in the middle of the immersion and wrong at both ends.** Within 10 % of the photograph over the middle third; too short within a third of a chord of the surface; absent over the lower quarter because of the one-panel hold criterion.

Two changes follow, neither made yet so that the results above stay reproducible: shed from the tip only on a strut, and replace the one-panel hold criterion by one on the sectional length. Sects. 6 to 9 should then be rerun. Two things could not be checked: the Froude number of Harwood's lifting-line lift curves was not recorded when they were digitised and is taken as 1.5, and the figure numbers of the source paper are not recorded in the data files.

10.7 Digitising the remaining data

The items of Table 10.1 marked "to digitise" follow the same route as the four done. Obtain the figure at the highest resolution available, the publisher's PDF rather than a screen capture, and check first whether the data exist as tables: the authors of the 2026 paper are likely to share theirs. Use WebPlotDigitizer (<https://automeris.io>) or an equivalent, calibrate both axes on four known ticks far apart, and extract each series separately, one regime or one Froude number at a time, recording the marker type because in these papers it encodes the regime or the path. Take the uncertainty as half the marker size in data units, or the error bar where one is drawn. Check the result by re-plotting the CSV over the original figure and by comparing two or three values with any quoted in the text. Two items are not curves but are needed alongside: the coordinates of Harwood's modified NACA 0009 section, including its trailing-edge thickness, so that the section table is built for the section tested; and the chord and immersion of each campaign, since the Reynolds number enters the separation criterion.

Keep the raw digitised files in `docs/validation/data/raw/` and add a line to its README stating their source. Convert them with `import_digitised.py` into one CSV per source and quantity in `docs/validation/data/`, with the columns of `TEMPLATE.csv`:

```
source, figure, quantity, regime, path, alpha_deg, Fn_h, AR_h, chord_m, section, value, uncertainty, notes
```

quantity is one of `CL`, `CD`, `CM`, `L_over_c`, `phi_deg`, `alpha_incep_deg`, `Fn_h_washout`, or a sectional quantity of a reference model (`CL_section`, `a0_over_2pi`, `alpha_eff_over_alpha`, `sigma_c`); regime is `FW`, `PV` or `FV`; path is `steady`, `increasing`, `decreasing`, `accelerating` or `perturbed`; a depth coordinate goes in `notes` as `z_over_h=...`. For a wing without free surface `Fn_h` is empty and `AR_h` holds the aspect ratio. Add a case to `validate.py` following the pattern of Cases A to D. A comparison is a success when the code lies within the digitisation uncertainty plus the scatter between repeat runs of the experiment, where repeats exist. Where it does not, the difference is the result: record it with the mechanism you believe explains it, as Sects. 10.3 to 10.5 do.

11 Limitations

- The free surface is the high-Froude image; results below $Fn_c = 1.5$ are not validated. The measured wetted lift slope falls by 13 % between $Fn_h = 1.5$ and 3.5 at $AR_h = 1$ (Sect. 10.3), a Froude effect the image cannot represent.
- Vortex shedding from the waterline edge of a strut keeps a finite loading at the surface and raises the wetted lift by 8 % above the high-Froude measurements; shedding from the tip only removes the excess (Sect. 10.3). Tip shedding on a swept wing with cosine spacing diverges at low incidence and must be off (Sect. 10.2).
- The boundary layer is in the section table, not in the lattice. Weber-number effects at model scale are not represented.
- The cavity length is sectional; the lattice adds the three-dimensional coupling of the loads but not of the cavity shape. The closure angle is 15 % low at the one published condition. The cavity is too short within a third of a chord of the surface and, because a section whose length is below one chordwise panel is declared unable to hold a cavity, absent over the lower quarter of the immersion at $\alpha = 10^\circ$, $Fn_h = 1.5$ (Sect. 10.5).
- The tip-vortex route and the tail base pressure are unvalidated.
- Wetted drag, cavity drag and spray are not modelled.

- The formation rate and the washout time are parameters, chosen from the 3.5 to 7 convective times of Aguiar Ferreira et al. and not fitted.
- The pressure-difference load routine inherited from the internship runs but disagrees with the Kutta–Joukowski loads by 40 % and should not be used.

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